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Hybrid aggregation strategy with double inverted residual blocks for lightweight salient object detection

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ABSTRACT

Lightweight salient object detection (SOD) is widely used in various downstream applications due to its low resource requirements and fast inference speed. The use of hybrid encoders offers the potential to achieve a better balance between efficiency and accuracy for SOD task. However, the aggregation of features from convolutional neural networks (CNNs) and transformers remains challenging, and most existing lightweight SOD models rarely explore the efficient aggregation of cross-architecture features derived from hybrid encoders. In this paper, we propose a hybrid aggregation strategy network (HASNet) that balances accuracy and efficiency for lightweight SOD by grouping and aggregating features to leverage salient information across different architectures. Specifically, the features obtained after hybrid encoder processing are divided into convolutional and transformer features for shallow and deep aggregation respectively. Deep aggregation uses the global inverted residual block (GIRB) to facilitate the transfer of salient information encoded within transformer features across various levels. Meanwhile, shallow aggregation uses the lightweight inverted residual block (LIRB) to efficiently integrate the spatial information inherent in convolutional features. The GIRB incorporates an efficient global operation to extract channel semantic information from the high-dimensional transformer features. The LIRB fuses low-level features by efficiently exploiting the spatial information in features at extremely low computational cost. Comprehensive experiments conducted across five datasets demonstrate that our HASNet significantly outperform existing methods in a thorough evaluation encompassing parameter sizes, inference speed, and accuracy. The source code will be publicly available at <https://github.com/LitterMa-820/HASNet>.

1. Introduction

Salient object detection (SOD) aims to extract the most attractive parts of images and plays a crucial role in various downstream tasks, including object tracking (Wang et al., 2020a; Yan et al., 2020), image annotation (Zhou et al., 2019), image smoothing (Zhu et al., 2024), image segmentation (Xu et al., 2023; Zhou et al., 2023), and image compression (Wang et al., 2020b).

Existing high-performance SOD methods require significant computational resources and are unsuitable for mobile and embedded device application scenarios. Additionally, a substantial gap exists between current lightweight and high-performance SOD approaches. As shown in Fig. 1, heavyweight SOD methods achieve superior performance compared to lightweight ones due to their larger parameter sizes, but their inference speed is significantly constrained. Convolutional neural networks (CNNs) architectures are superior at processing local fea-

tures, such as edges and textures. However, their limited receptive field makes it difficult for them to understand global contextual information. Improving the performance of SOD based on CNNs architectures requires multiple stacked convolutional layers, which significantly increases the number of parameters. The transformer architecture utilizes a self-attention mechanism to understand the overall structure of an image and establish long-distance dependencies. However, the high cost of matrix multiplication and the large number of parameters required for linear projection also lead to existing high-performance SOD being heavyweight. As illustrated in Fig. 2, we exhibit three common SOD architectures, labeled (a)-(c), along with the hybrid architecture used in our approach (d). CNN-based models like (a) (Pang et al., 2020; Qiu et al., 2022a; Tang et al., 2021; Wu et al., 2021) are limited by a restricted receptive field, often resulting in suboptimal performance in lightweight SOD. In contrast, SOD methods based on transformer architecture like (b) (Chen et al., 2024; Yun & Lin, 2024) improve

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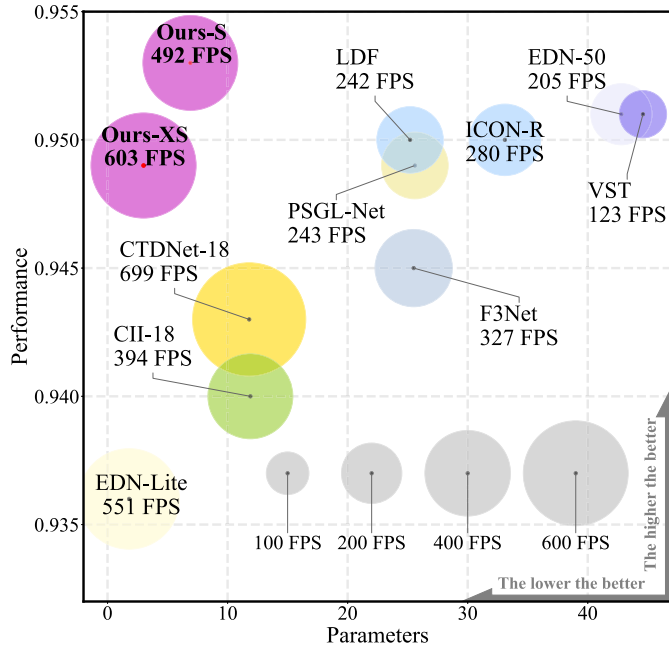


Fig. 1. Comparisons of our proposed method with other SOD models in accuracy, parameters and inference speed. The F-measure on the ECSSD dataset is used to evaluate the performance of the model. The size of the colored circles indicates the frames per second (FPS) of each method, with larger circles representing higher inference speeds.

segmentation accuracy but demand higher computational resources. Combining the local perceptual capabilities of CNNs with the global semantic information of transformers like (c) (Ma et al., 2023; Qiu et al., 2024; Song et al., 2024) enhances both accuracy and imaging quality, though it introduces significant parameter and computational overhead. To summarise the above analysis, we combine the advantages of CNNs and transformer, adopt a single hybrid encoder as the backbone of SOD, and design a hybrid aggregation strategy (HAS) for cross-architecture features to minimise computational cost while improving accuracy, as shown in (d). However, due to differences in the focus on spatial and channel information between CNNs and transformer features, effectively aggregating salient information from both is challenging. HAS comprises three sub-aggregation steps: transformer feature aggregation for global semantic information enhancement, CNNs feature aggregation for local spatial information enhancement, and hybrid feature aggregation for cross-architecture information fusion. For transformer feature aggregation, we designed a global inverted residual block (GIRB) that deeply aggregates high-level channel semantic features generated by the multi-head self-attention mechanism. For CNNs and cross-architecture feature aggregation, we designed a lightweight inverted residual block (LIRB) that combines spatial and channel information from both architectural features through effective channel and spatial transformations.

The inverted residual block (IRB) (Sandler et al., 2018) operates on the spatial and channel dimensions of the features, and is expected to

process feature information from both architectures. However, it lacks optimisation of the attention mechanism for transformer feature aggregation and excessive linear transformations are expensive for CNNs feature aggregation. For transformer feature aggregation, we design GIRB to embed cross-scale attention as an efficient global operation into IRB. GIRB combines information from two neighbouring scales, enhancing their joint representation in channel semantics. Specifically, the GIRB improves the efficiency of attentional computation by ignoring a small number of interactions between patches to aggregate information from two neighbouring features. However, as GIRB focuses more on channel information relating to high-level semantics, the folding operation does not significantly degrade performance. In cross-architecture and CNNs feature aggregation, two 1×1 convolutions in IRB will focus too much on the channel information resulting in channel redundancy and waste of computational resources. Inspired by some lightwighting approaches (Chen et al., 2023; Mehta & Rastegari, 2021), we improve the two convolutions in IRB into expansion stage and shrinkage stage by introducing identity splicing and channel merging. In the expansion stage, we halve the number of 1×1 convolutional kernels and add identity branches to realize channel expansion. The reduced number of kernels provides channel information processing capability, while the number of spliced channels maintains spatial feature diversity. After depth-wise convolution processing, the expanded features have similar spatial information in neighbouring channels. Thus, two neighboring channels are weighted and summed in the shrinkage stage to condense the salient information of the features. Ultimately, the two stages designed in LIRB greatly reduce both the size of the parameters and the number of Giga floating-point operations per second (GFLOP) induced by 1×1 in IRB.

In summary, we propose a novel hybrid feature aggregation strategy network (HASNet) for lightweight SOD. HASNet combines the local modeling capabilities of the CNNs architecture with the global modeling capabilities of the transformer architecture, achieving a balance between efficiency and performance. Additionally, we design LIRB and GIRB to optimize the features from different architectures during the aggregation and decoding processes. By combining HAS, LIRB and GIRB, the HASNet achieves higher saliency detection performance than existing methods, and the smaller image input size requirement makes the model less computationally expensive and faster to infer. As shown in Fig. 1, HASNet achieves significant improvements in performance and efficiency compared to existing SOD methods.

- We propose a feature aggregation strategy called HAS to efficiently utilize and integrate feature information from different architectures. This strategy groups features from various architectures and processes them through shallow and deep aggregation to integrate information from adjacent levels.
- We propose GIRB and LIRB to extract information from different architectural features during HAS and decoding. In GIRB, the introduction of efficient global operation enhances the ability to capture channel information for transformer features. In LIRB, expansion and shrinkage steps replace linear transformations in IRB, significantly reducing parameter size and computational cost.

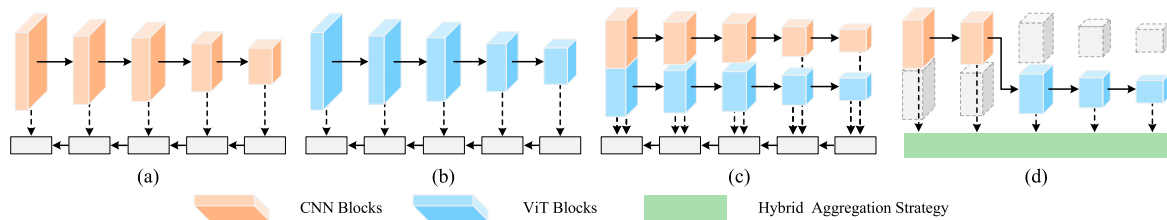


Fig. 2. Illustration of different feature encoders. Orange blocks indicate convolution modules and blue blocks indicate transformer modules. (a) Methods using a fully convolutional encoder. (b) Methods with full transformer encoder. (c) Methods using both convolutional and transformer dual encoders. (d) Our multi-architecture hybrid single encoder and hybrid feature aggregation approach.

- Extensive experiments show that our HASNet, which combines the advantages of CNNs and transformer architectures, outperforms SOTA methods in both quantitative and qualitative evaluations on five datasets.

2. Related works

2.1. Heavyweight salient object detection

The task of SOD is to extract the most visually significant regions from an image. In general, SOD methods that use convolutional (He et al., 2016; Simonyan & Zisserman, 2014) or transformer (Dosovitskiy et al., 2021; Ranftl et al., 2021; Wang et al., 2021; Wu et al., 2023) architectures with a large number of parameters as a backbone are referred to as heavyweight SOD. Their primary optimization focus is not on inference speed or resource efficiency but rather on achieving substantial improvements in performance. At the beginning, researchers are exploring SOD on the basis of purely convolutional architectures. Su et al. (2019) proposed a feature-selective network with a boundary-aware component that enhances feature selectivity at the boundaries by merging boundary localization streams, while ensuring feature invariance in the interior through a complex, interior-aware stream. Qiu et al. (2022b) proposed atrous spatial pyramid pooling to impose preferences on various feature scales. Li et al. (2021) proposed a novel convolutional neural network with purification mechanism and loss of structural similarity to solve the problem of indistinguishable foreground and background. The advent of the transformer architecture has led to greater performance gains in SOD, but at a higher computational cost. Li et al. (2023) proposed GeleNet to initially employ a transformer backbone to generate four-level feature embeddings with global, long-range dependencies. Multi-tasking and knowledge distillation were also used to enhance the performance of SOD. For example, Liu et al. (2024, 2021b) developed a multi-task transformer decoder that simultaneously predicts significance and boundary results within pure transformer architectures. Liu et al. (2023b) proposed a scene-guided dual-branch network that performs cross-task knowledge distillation from a scene classification task to facilitate saliency detection. Deep supervised training is also a common method to improve the performance of SOD. Huo et al. (2024) fed multimodal features into triplet decoder and used hierarchical deep supervised loss to further enable the network to capture unique saliency cues.

In summary, the heavyweight SOD model has achieved impressive segmentation accuracy. However, both convolution and transformer architectures struggle to effectively balance inference speed and performance, significantly limiting their applicability to downstream tasks.

2.2. Lightweight salient object detection

The advent of lightweight SOD models has helped alleviate this problem and has produced encouraging results. SAMNet proposed by Liu et al. (2021c) introduces a novel stereo attention multi-scale module, which uses a stereo attention mechanism to adaptively fuse features of various scales. Liu et al. (2020) also proposed a hierarchical visual perception module in HVPNet to imitate the primate visual cortex for hierarchical perceptual learning. Wang et al. (2024, 2023c) proposed two extremely lightweight SOD methods, ELWNet and LARNet, which eliminate the dependence on ImageNet pre-training. Some researchers have used the MobileNet family (Howard et al., 2019, 2017; Sandler et al., 2018) being used as a backbone for lightweight SOD. Zhao et al. (2021) proposed an efficient and effective complementary trilateral decoder (CTD) for saliency detection with three branches: semantic path, spatial path, and boundary path. Liu et al. (2021a) proposed a centralized information interaction (CII) strategy and a relative global calibration module, which only introduced a small amount of additional parameters and FLOPs to achieve superior results. Some lightweight detection methods have also emerged in similar fields. For example,

Huang et al. (2023) proposed a novel lightweight distortion-aware network in the field of omnidirectional image SOD. Zeng et al. (2024) introduced AirSOD, which utilizes the first three layers of MobileNetV2 features and captures both remote dependencies and local information for RGB-D SOD through parallel attention-shift convolution. In terms of lightweighting techniques for RGB-T SOD, Huo et al. (2021) adopted a lightweight backbone and designed efficient decoder parts, which greatly reduced the computational cost. OSRNet (Huo et al., 2022) is designed as a one-stream network to avoid the increased computational burden of the dual-stream encoder structure commonly used in RGB-T SOD. Shen et al. (2024) designed a multi-scale interaction module for real-time SOD of surface defects on steel strips, utilizing depth-wise convolution and pointwise convolution to independently extract and interactively fuse features at different scales. Liu et al. (2023a) designed a multi-task based super-resolution assisted learning framework to enhance the performance of SOD in optical remote sensing images (RSI-SOD) with low-resolution inputs.

Although these lightweight methods achieve feasible segmentation results with good inference speed, they still fall short of high-performance SOD. Additionally, these approaches seldom explore effective integration of transformer architectures and feature aggregation across different architectures. In response, our approach introduces HAS, achieving highly accurate salient object detection with efficient inference speed.

2.3. Hybrid encoder

Recent research has introduced hybrid architecture encoders to improve the segmentation performance and accuracy of SOD. The MobileViT family (Mehta & Rastegari, 2021, 2022; Wadekar & Chaurasia, 2022), the first two layers are MV2 consisting of convolutions, and the last three layers are transformer architecture. There have also been many studies (Wang et al., 2022b, 2021; Wu et al., 2023) embedding convolutional operations into the transformer structure to achieve a combination of local and global receptive fields. In segmentation tasks, some researchers have employed a dual encoder structure combining CNNs and transformer architectures to enhance model performance. For instance, Yuan et al. (2023) proposed a complementary network integrating CNNs and transformer for medical image segmentation. Song et al. (2024) proposed a two-branch encoder by combining P2T and ResNet to ensure that the extracted features contain both rich global semantic information and local detailed features. Qiu et al. (2024) proposed a converter-based asymmetric bilateral U-Net by combining the transformer architecture and U-Net. Wang et al. (2022a) employed the combination of the first three layers of ResNet50 and the last two layers of PVTv2 as a single encoder to extract rich information in the shallow and deep stages for RSI-SOD. Some researchers have also enhanced the representation of features by adding transformer modules after the CNNs architecture encoder. Liu et al. (2021d) proposed to make the convolutional features go through a set of shared transformer modules so that features at different levels express similar semantics.

Previous approaches have attempted various strategies to combine CNNs and transformer architectures, but these methods primarily focus on heavyweight SOD. In this paper, we propose a hybrid feature aggregation strategy across architectures to develop a lightweight SOD approach that balances performance and efficiency.

3. Method

3.1. Overview

The overall architecture of our HASNet is shown in Fig. 3. A given input image of size 256×256 is encoded by the MobileViT backbone network to obtain five layers of features $X_{1 \sim 5}$ from different scales. The first two layers in MobileViT consist solely of CNNs architectures, while the last three layers are entirely composed of transformers. Therefore,

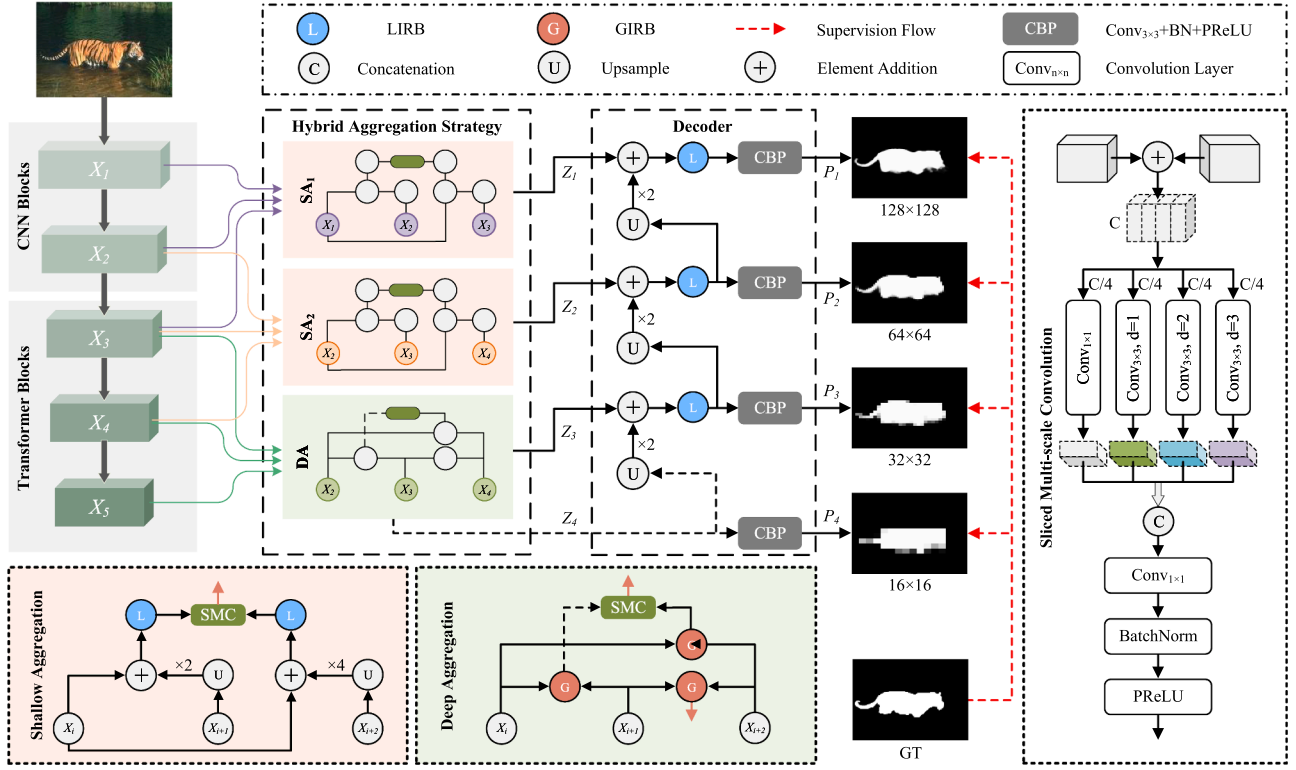


Fig. 3. Overall architecture of our HASNet. A MobileViT with a mix of CNNs and transformer modules is used as an encoder. A feature grouping strategy consisting of two shallow aggregations and one deep aggregation is used to process features at all levels. The aggregated features are processed through a decoder consisting of LIRB and CBP to get one trunk output and three branch outputs.

we categorize the features generated by the two different architectures, with $X_{1\sim2}$ classified as CNNs features containing more spatial geometrical information, and $X_{3\sim5}$ classified as transformer features with richer channel information. Subsequently, three group aggregation steps in the HAS aggregate features into a total of four features $Z_{1\sim4}$ to ensure network depth while reducing computational cost. Then, a decoder consisting of a series of LIRBs fuses the multi-scale features from top to bottom and four CBPs convert the features into saliency maps $P_{1\sim4}$ at four different scales. Finally, we use the ground truth for supervised training of $P_{1\sim4}$ at different scales and use P_1 as the main output of our method.

3.2. Hybrid aggregation strategy

The features obtained from different coding architectures in the hybrid encoder focus on different salient information. We propose a hybrid aggregation strategy to group and aggregate these features. Specifically, we divide the features into two categories: CNNs features $X_{1\sim2}$, and transformer features $X_{3\sim5}$. These two types of features are mixed into three groups participating in two shallow aggregations (SA_{1~2}) and one deep aggregation (DA) respectively. SA consists mainly of LIRBs, which can aggregate spatially represented CNNs features at a low computational cost and transform deep semantic features into spatial representations through linear transformations. The DA consists of the GIRD and is responsible for aggregating the deep features of the transformer architecture. The efficient global operation embedded within it further integrates the channel information of features across different scales. In addition, we designed a sliced multi-scale convolution (SMC) to further fuse the aggregated same-size features in the final part of the feature aggregation process in both SA and DA.

At the start, the features $X_{1\sim3}$ and $X_{2\sim4}$ are fed into the two shallow aggregation steps of the HAS. In SA, we design a pyramid-structured upward aggregation method to fuse the small-scale features with the largest-scale features separately and finally merge all the features. In this

way, we can maintain the integrity of the imaging as much as possible. Specifically, $X_{i+1\sim i+2}$ is upsampled and added with X_i , and then the results are aggregated by LIRB to obtain $X_{i,i+1}$ and $X_{i,i+2}$, which are expressed as follows:

$$\begin{aligned} X_{i,i+1} &= \text{LIRB}_{i,i+1}(X_i + \text{Upsample}(X_{i+1})) \\ X_{i,i+2} &= \text{LIRB}_{i,i+2}(X_i + \text{Upsample}(X_{i+2})) \end{aligned} \quad (1)$$

where $\text{Upsample}(\cdot)$ represents bilinear interpolation upsampling, $\text{LIRB}_{i,j}$ represents the LIRB module that aggregates X_i and X_j .

$X_{i,i+1}$ and $X_{i,i+2}$ get the same shape in the aggregation process, and we further obtain $Z_{1\sim2}$ by aggregating their salient information through SMC. The process is represented as follows:

$$Z_i = \text{SMC}_1(X_{i,i+1}, X_{i,i+2}) \quad (2)$$

The $X_{3\sim5}$ are transformer architecture features with high-level channel semantic information, and we designed GIRD to aggregate these features in DA. Firstly, $X_{4\sim5}$ and X_3 are aggregated by GIRD to obtain X_{34} and X_{35} , respectively, as follows:

$$\begin{aligned} X_{34} &= \text{GIRD}_{34}(X_3, X_4) \\ X_{35} &= \text{GIRD}_{35}(X_3, X_5) \end{aligned} \quad (3)$$

where, $\text{GIRD}_{i,j}$ represents the GIRD module that aggregates X_i and X_j .

Then, we use SMC to aggregate X_{34} and X_{35} on the same scale to obtain Z_3 . In addition, in order to enhance the feature depth of the model output, we process X_4 and X_5 through GIRD to obtain Z_4 , as follows:

$$\begin{aligned} Z_3 &= \text{SMC}_3(X_{34}, X_{35}) \\ Z_4 &= \text{GIRD}(X_4, X_5) \end{aligned} \quad (4)$$

After all the feature aggregation operations, we obtain four features $Z_{1\sim4}$.

It is also worth noting that the role of SMC in Eq. (2) is to slice the input feature x into $x_{1\sim4}$ by the channel dimension, where x_1 uses

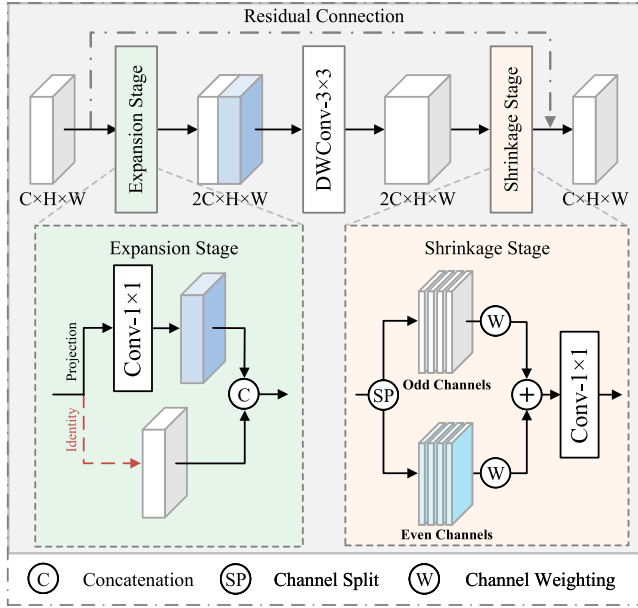


Fig. 4. The architecture of LIRB.

$Conv_{1 \times 1}$ to enhance the channel dimension information, and $x_{2 \sim 4}$ undergoes dilated convolution with different dilation rates to search for spatially salient information. The output x' is obtained by aggregating the four branch slices through a set of $Conv_{1 \times 1}$ and concatenation operations. The whole process can be shown as follows:

$$\begin{cases} x_{1 \sim 4} = Slice(x) \\ x'_1 = Conv_{1 \times 1}(x_1) \\ x'_{2 \sim 4} = Conv_{3 \times 3}^{d_{1 \sim d_3}}(x_{2 \sim 4}) \\ x' = Conv_{1 \times 1}(Cat(x'_{1 \sim 4})) \end{cases} \quad (5)$$

where $Slice(\cdot)$ means to slice x into 4 equal parts along the channel dimension, $Conv_{n \times n}$ denotes the convolution operation with a kernel size of $n \times n$, $Conv_{3 \times 3}^{d_{1 \sim d_3}}$ denotes three 3×3 convolutions with dilation ratios of 1 ~ 3 and $Cat(\cdot)$ denotes concatenation along the channel dimension.

3.3. Lightweight inverted residual block

The design of depth-wise convolution in the IRB is well-suited for aggregating spatially oriented convolutional features, whereas linear projection is suitable for exploiting channel information in transformer features. However, the traditional IRB has some shortcomings. First, the 1×1 convolution responsible for expanding and shrinking the number of feature channels consumes a large number of parameters and GFLOPs. Second, the excessive number of channels in the channel expansion stage leads to feature redundancy. Thus, we propose a lightweight inverted residual block (LIRB) to improve both the expansion and shrinkage stages of the traditional IRB, as shown in Fig. 4. Unlike ShuffleNet (Ma et al., 2018; Zhang et al., 2018), where the computational complexity of the 1×1 convolution is reduced by channel shuffling operations, our main goal is to efficiently implement the increase and decrease of the number of feature channels. During the expansion stage, we merge the identity and projection branches to increase the number of feature channels. In the shrinking stage, we merge redundant information between neighbouring channels. In addition, we projected the number of feature channels to $2 \times$ instead of $4 \times$ for conventional IRBs. After these two improvements, we significantly reduced redundant feature information, greatly enhancing the efficiency of the IRB.

To begin with, we use the expansion stage to replace the first linear projection layer in the IRB to minimize the waste of computational

resources for channel transformations, denoted as follows:

$$\begin{aligned} X' &= Conv_{1 \times 1}(X) \\ X^e &= Cat(X', X) \end{aligned} \quad (6)$$

where X^e is the feature after expanding the number of channels, which is obtained by channel concatenation operation of X' with the original input X . The X' is the feature after linear projection, which keeps the number of channels the same as X .

Consistent with the default IRB after the expansion phase, a 3×3 depth-wise convolution is performed as follows:

$$X^{e'} = DWConv_{3 \times 3}(X^e) \quad (7)$$

where $DWConv_{3 \times 3}$ denotes a depth-wise convolution with a convolution kernel size of 3×3 .

Then, we design the shrinkage stage to replace another linear projection layer for reducing the number of channels in the IRB. The given $X^{e'}$ is sliced into two parts in odd-even order, represented as $X_{\mu}^{e'}$ and $X_{\eta}^{e'}$, as follows:

$$X_{\mu}^{e'}, X_{\eta}^{e'} = SSlice(X^{e'}) \quad (8)$$

where $SSlice(\cdot)$ represents the odd-even interleaved slicing operation, $X_{\mu}^{e'}$ and $X_{\eta}^{e'}$ are the odd and even parts of $X^{e'}$ respectively.

Neighboring feature maps exhibit high similarity, and overlapping them helps reduce information redundancy. Therefore, we merge the two slices after assigning weights to the channel slices. This process further reduces unimportant information in the channel as follows:

$$X^s = \gamma_{\mu} \cdot X_{\mu}^{e'} + \gamma_{\eta} \cdot X_{\eta}^{e'} \quad (9)$$

where γ_{μ} and γ_{η} represent the learnable parameters for half the number of channels of $X^{e'}$.

Finally, a 1×1 convolution projects the channel-reduced features, complementing the linear transformation performed during the shrinkage stage, which is formulated as:

$$X^{s'} = Conv_{1 \times 1}(X^s) \quad (10)$$

where $X^{s'}$ is the result obtained after processing all the steps in the aforementioned LIRB.

3.4. Global inverted residual block

Conventional IRBs include only simple linear transformations of channels and are unable to effectively aggregate multi-head self-attention features encoded by the Transformer architecture. Therefore, we design GIRB based on IRB to enhance the semantic representation when aggregating the multi-head self-attention feature $X_{3 \sim 5}$ through efficient global operations. Compared to the conventional multi-head self-attention operations, the efficient global operation of the GIRB leverages the complementary semantic hierarchies of neighboring features to enhance global understanding without incurring excessive computational costs. As shown in Fig. 5, GIRB aggregates information from two neighboring scale features using an attention operation with a folding operation. Optimisation is then performed by convolutional blocks in the channel and spatial dimensions, respectively.

To start, we define the two inputs to the GIRB as X_L and X_H , which have the following shapes: $C_1 \times H_1 \times W_1$ and $C \times H_2 \times W_2$. To ensure that both features have the same number of channels, a 1×1 convolution is used to convert the number of channels in X_L into C . We reduce the number of patches interacting in efficient global operations by folding patches as follows:

$$\begin{aligned} X_L^f &= fold(Conv_{1 \times 1}(X_L)) \\ X_H^f &= fold(X_H) \end{aligned} \quad (11)$$

where $fold(\cdot)$ denotes the folding operation, and X_L^f and X_H^f are the folded features, respectively. Similar to the way (Wang et al., 2022a) folds features into the channel dimension, we fold them into the batch

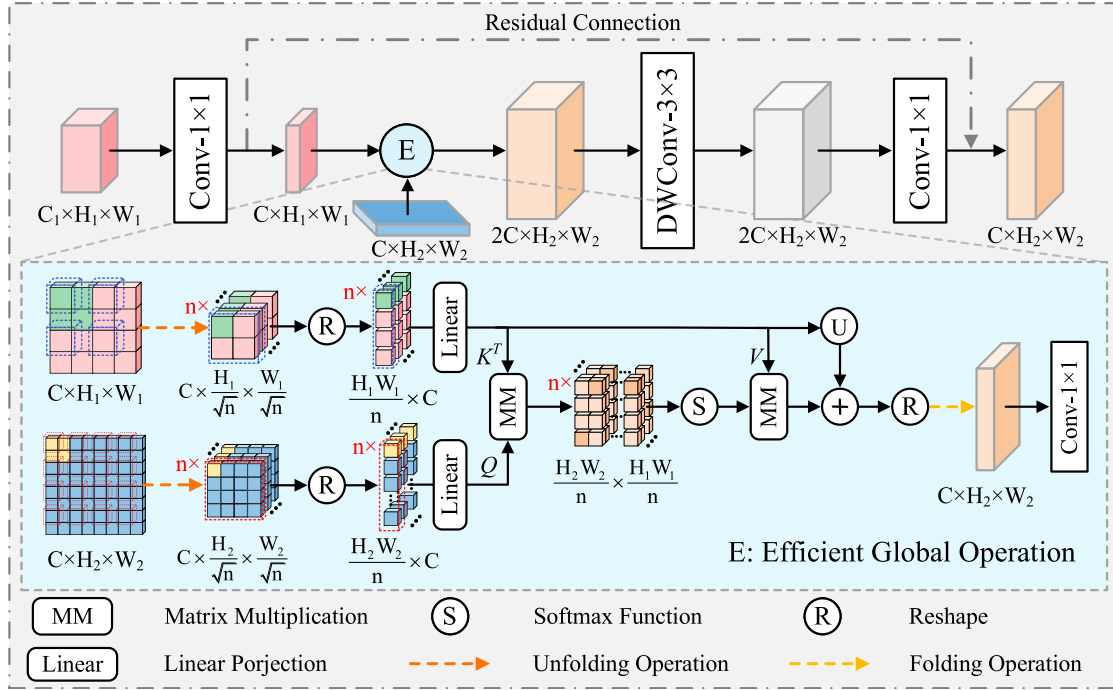


Fig. 5. The architecture of GIRB. K^T is denoted as the transpose of K .

dimension. Following the folding operation, the shapes of the two features change from $C \times H_1 \times W_1$ to $n \times C \times \frac{H_1}{\sqrt{n}} \times \frac{W_1}{\sqrt{n}}$, and from $C \times H_2 \times W_2$ to $n \times C \times \frac{H_2}{\sqrt{n}} \times \frac{W_2}{\sqrt{n}}$.

In subsequent attention operations, each folded patch is computed only with the corresponding $\frac{H_i \times W_i}{n}$ patches in another feature. The folding operation probably causes interaction losses in the patches. Using the window to compute these patches in subsequent convolutional blocks will attempt to replenish this information. Specifically, X_L^f and X_H^f are reshaped to $n \times \frac{W_1 H_1}{n} \times C$ and $n \times \frac{W_2 H_2}{n} \times C$. A set of linear projections converts them to Q , K and V as follows:

$$\begin{aligned} Q &= \text{proj}_q(\text{reshape}(X_H^f)) \\ K, V &= \text{proj}_{k,v}(\text{reshape}(X_L^f)) \end{aligned} \quad (12)$$

where $\text{proj}_q(\cdot)$ and $\text{proj}_{k,v}(\cdot)$ denote linear projection operations. Linear projection will not change the shape of X_L^f and X_H^f .

Q and K^T are matrix multiplied to obtain an attention weight matrix of shape $n \times \frac{W_2 H_2}{n} \times \frac{W_1 H_1}{n}$. This attention weight matrix is multiplied with V to obtain the shape $n \times \frac{W_2 H_2}{n} \times C$ which is then summed with the residuals. After reshaping and unfolding, the feature X_g with dimensions $C \times H_2 \times W_2$ is obtained, which is expressed as:

$$\begin{aligned} X_g^f &= \delta \left(\frac{QK^T}{\sqrt{D/N}} \right) V + \text{Upsample}(X_L^f) \\ X_g &= \text{unfold}(\text{reshape}(X_g^f)) \end{aligned} \quad (13)$$

where δ is the Softmax activation function. D denotes the number of channels of Q , K and V , and N denotes the number of attention heads in the multi-head attention mechanism. $\text{Unfold}(\cdot)$ is the inverse of the folding operation, restoring x_g^f to its original dimensions.

After the unfolding operation to recover the feature shape, we use the 1×1 convolution and the 3×3 depth-wise convolution to complement the patch interaction information lost in the same folded window, as follows:

$$X_g' = \text{Conv}_{1 \times 1}(\text{DWConv}_{3 \times 3}(\text{Conv}_{1 \times 1}(X_g))) \quad (14)$$

3.5. Loss function

We designed a decoder to convert the four features $Z_1 \sim Z_4$ obtained after the aggregation step into four saliency maps at different scales, as follows:

$$\begin{aligned} Z_i' &= \text{LIRB}_i(Z_i + \text{Upsample}(Z_{i+1})) \\ P_i &= \text{CBP}_i(Z_i') \end{aligned} \quad (15)$$

where CBP_i consists of a convolution with a 3×3 kernel size, Batch-Norm, and the PReLU activation function. P_i represents the four outputs of our method for supervised learning.

The weighted binary cross entropy ($\mathcal{L}_{bce}^w$) and weighted IoU ($\mathcal{L}_{iou}^w$) loss functions are used for the supervised learning of all branching outputs, $P_{1 \sim 4}$, and are defined as follows:

$$\mathcal{L}_i = \mathcal{L}_{bce}^w(GT_i, P_i) + \mathcal{L}_{iou}^w(GT_i, P_i) \quad (16)$$

where GT_i is the ground truth corresponding to the resolution of P_i and $\mathcal{L}_i$ denotes the loss between P_i and GT_i .

We adopt the sum of all branch losses $\mathcal{L}_{sod}$ as the loss function during training, expressed as follows:

$$\mathcal{L}_{sod} = \sum_{i=1}^4 \mathcal{L}_i \quad (17)$$

4. Experiment

4.1. Datasets

We evaluate our proposed models on the five most well-known SOD datasets: DUTS (Wang et al., 2017), ECSSD (Yan et al., 2013), HKU-IS (Li & Yu, 2015), PASCAL-S (Li et al., 2014), and DUT-OMRON (Yang et al., 2013). DUTS contains 10,553 training images and 5019 test images. ECSSD includes 1000 images, HKU-IS contains 4447 images, PASCAL-S has 850 images, and DUT-OMRON consists of 5168 images. In line with most methods, we use the 10,553 images in the DUTS training set to train our model and validate it using the other datasets.

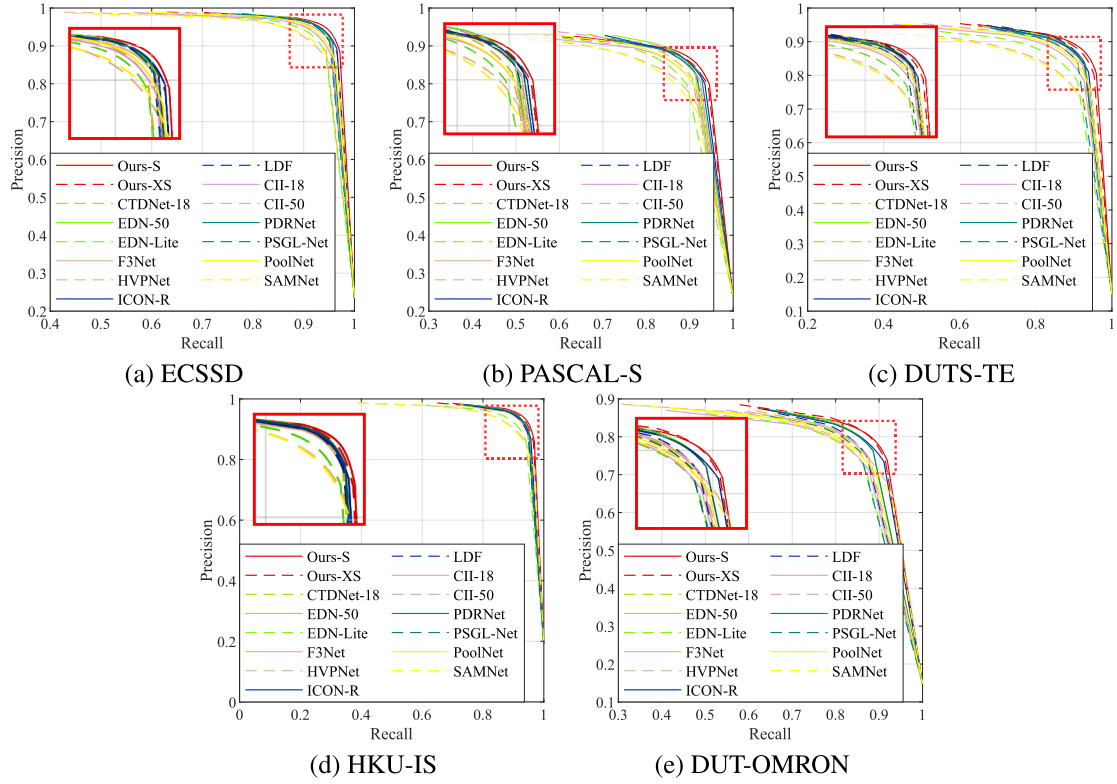


Fig. 6. PR curves comparison of different models on five SOD datasets. The solid and dotted lines represent Ours-XS and Ours-S, respectively.

Table 1

Performance comparison of our methods with 16 SOTA methods, including 9 lightweighting methods. * indicates that the method is tested using a batch size of 8. $\downarrow$ (↑) means that the higher (Lower) is better. $F_{\beta}^{\max}$ and S_{α} denote the F-measure and S-measure, respectively. The best and second best lightweight results are shown in **bold** and *italic*.

Method	#Pub (Year)	Params (M)	FLOPs (G)	Speed (FPS)	ECSSD			PASCAL-S			DUTS-TE			HKU-IS			DUT-OMRON		
					$F_{\beta}^{\max}$	MAE	$S_{\alpha} \uparrow$	$F_{\beta}^{\max}$	MAE	$S_{\alpha} \uparrow$	$F_{\beta}^{\max}$	MAE	$S_{\alpha} \uparrow$	$F_{\beta}^{\max}$	MAE	$S_{\alpha} \uparrow$	$F_{\beta}^{\max}$	MAE	$S_{\alpha} \uparrow$
					$\uparrow$	$\downarrow$		$\uparrow$	$\downarrow$		$\uparrow$	$\downarrow$		$\uparrow$	$\downarrow$		$\uparrow$	$\downarrow$	
Heavyweight Methods																			
PoolNet	CVPR ₁₉	68.3	88.9	95	0.944	0.039	0.921	0.863	0.074	0.843	0.880	0.040	0.883	0.933	0.032	0.917	0.808	0.056	0.836
F3Net	AAAI ₂₀	25.5	16.5	327	0.945	0.033	0.924	0.872	0.061	0.854	0.891	0.035	0.888	0.937	0.028	0.917	0.813	0.0553	0.838
LDF	CVPR ₂₀	25.2	15.6	342	0.950	0.034	0.924	0.874	0.059	0.856	0.897	0.034	0.892	0.939	0.027	0.919	0.820	0.051	0.838
CII-50	TIP ₂₁	24.5	21.9	260	0.950	0.033	0.926	0.877	0.061	0.858	0.888	0.037	0.887	0.939	0.029	0.920	0.816	0.054	0.839
VST	CVPR ₂₁	44.6	23.2	123	0.951	0.033	0.932	0.875	0.061	0.865	0.890	0.037	0.896	0.942	0.029	0.928	0.825	0.058	0.850
PSGL-Net	TIP ₂₁	25.6	19.5	243	0.949	0.031	0.925	0.873	0.061	0.854	0.886	0.036	0.883	0.938	0.027	0.919	0.811	0.052	0.831
PDRNet	TCSVT ₂₂	47.7	24.6	-	0.950	0.032	0.927	0.873	0.060	0.858	0.892	0.035	0.891	0.941	0.027	0.924	0.826	0.052	0.846
EDN-50	TIP ₂₂	42.8	20.4	205	0.951	0.032	0.927	0.880	0.061	0.858	0.894	0.035	0.892	0.941	0.026	0.924	0.828	0.048	0.846
MMFT	TIP ₂₂	72.1	25.5	-	0.960	0.029	0.936	0.894	0.059	0.871	0.904	0.034	0.893	0.944	0.029	0.936	0.831	0.049	0.849
ICON-R	TPAMI ₂₃	33.1	21.0	280	0.950	0.032	0.929	0.876	0.063	0.855	0.892	0.037	0.888	0.939	0.029	0.920	0.825	0.057	0.844
MENet*	CVPR ₂₃	27.8	94.7	57	0.955	0.031	0.928	0.890	0.053	0.865	0.913	0.028	0.905	0.948	0.023	0.960	0.834	0.045	0.850
BBRF	TIP ₂₃	74.5	46.4	117	0.963	0.022	0.939	0.891	0.049	0.871	0.916	0.025	0.908	0.952	0.020	0.932	0.843	0.042	0.855
ICON-S*	TPAMI ₂₃	92.4	52.8	83	0.961	0.023	0.941	0.896	0.048	0.878	0.920	0.025	0.917	0.951	0.022	0.935	0.855	0.042	0.869
Lightweight Methods																			
HVPNet	TCYB ₂₀	1.2	1.2	665	0.928	0.052	0.903	0.840	0.089	0.824	0.839	0.058	0.849	0.915	0.044	0.899	0.804	0.065	0.831
SAMNet	TIP ₂₁	1.3	0.6	589	0.928	0.050	0.907	0.839	0.090	0.820	0.836	0.058	0.848	0.915	0.045	0.898	0.803	0.065	0.830
CII-18	TIP ₂₁	11.9	16.0	394	0.940	0.039	0.916	0.860	0.067	0.845	0.876	0.043	0.874	0.932	0.032	0.912	0.801	0.058	0.828
CTDNet-18	MM ₂₁	11.8	6.2	699	0.943	0.037	0.917	0.869	0.065	0.848	0.885	0.037	0.882	0.939	0.028	0.918	0.817	0.052	0.837
EDN-Lite	TIP ₂₂	1.8	1.1	551	0.936	0.042	0.911	0.856	0.072	0.836	0.861	0.045	0.862	0.926	0.034	0.907	0.800	0.057	0.822
PoolNet-M+	TPAMI ₂₃	3.0	-	-	0.938	0.048	0.909	0.864	0.078	0.844	0.872	0.046	0.868	0.930	0.037	0.909	0.830	0.056	0.830
LSOD-P2T	CVIU ₂₄	12.3	2.7	-	0.942	0.037	0.922	0.862	0.067	0.857	0.872	0.041	0.879	0.931	0.033	0.915	0.816	0.058	0.841
ADMNet+	TMM ₂₄	0.8	3.3	418	0.925	0.049	0.900	0.828	0.087	0.809	0.840	0.052	0.849	0.918	0.036	0.901	0.798	0.058	0.826
SOLNet	TGRS ₂₅	6.5	8.1	469	0.855	0.079	0.828	0.767	0.122	0.748	0.724	0.087	0.758	0.848	0.061	0.835	0.701	0.089	0.752
HASNet-XS	-	3.0	0.5	603	<i>0.949</i>	<i>0.033</i>	<i>0.928</i>	<i>0.880</i>	<i>0.061</i>	<i>0.864</i>	<i>0.898</i>	<i>0.033</i>	<i>0.898</i>	<i>0.942</i>	<i>0.027</i>	<i>0.926</i>	<i>0.835</i>	<i>0.049</i>	<i>0.856</i>
HASNet-S	-	6.9	0.8	492	0.953	0.029	0.933	0.884	0.058	0.868	0.900	0.032	0.902	0.945	0.025	0.930	<i>0.832</i>	<i>0.052</i>	<i>0.854</i>

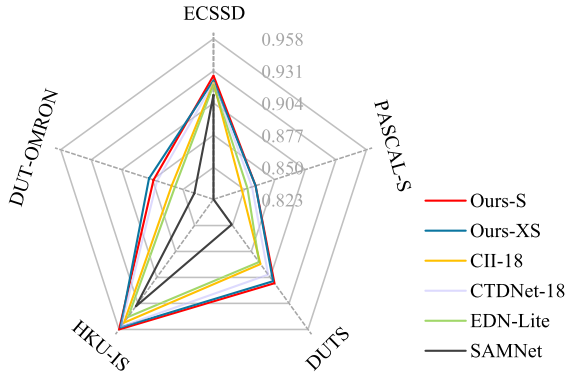


Fig. 7. Radar chart of E-measure metrics for our method compared to other lightweight methods.

4.2. Evaluation metrics

The SOD evaluation metrics used in this study include Precision-Recall (Borji et al., 2015), S-measure (Fan et al., 2017), F-measure (Achanta et al., 2009), E-measure (Fan et al., 2018), and Mean Absolute Error (MAE) (Perazzi et al., 2012). Precision-Recall represents the curve of precision and recall at various thresholds. Precision measures the accuracy of the detections, while recall measures the coverage of the ground truth objects. S-measure evaluates the structural similarity between the predicted map and the ground truth. It considers both the object region and the object structure. The F-measure combines precision and recall using a weighted harmonic mean. In this study, we use the max F-measure as the evaluation criterion. E-measure simultaneously evaluates global information and local pixel-wise matching information between the predicted map and the ground truth. MAE calculates the average absolute difference between the pixel values of the predicted map and the ground truth. MAE is one of the most important evaluation metrics in salient object detection. To evaluate the performance of our method, we use the evaluation toolkit consistent with Saliency-Evaluation-Toolbox¹ to compute these metrics.

4.3. Implementation details

Our experiments were conducted on a server configured with a 12th Gen Intel(R) Core(TM) i9-12900K CPU, 128 GB RAM, and a NVIDIA RTX 3090 (24 GB) GPU. The two models using MobileViT-XS and MobileViT-S as their backbones are named HASNet-XS and HASNet-S, respectively. The input size was set to 256×256 for images. We employed data enhancement techniques commonly used in SOD, including random flipping, rotation, and cropping to avoid overfitting. The backbone was initialized using the parameters of the classification tasks. The parameters of other parts of the model were initialized according to the default initialization scheme in PyTorch. Adam (Kingma & Ba, 2014) was used as the optimizer during training. The batch size was set to 16, and the initial learning rate was set to $5e-4$. The learning rate was decreased by a factor of 10 every 50 epochs. To speed up the training process and reduce resource consumption, we employed mixed precision training. Convergence was achieved within 100 epochs, taking approximately 2 h.

4.4. Comparison with SOTAs

In order to verify the efficiency of our method, we compare it with 16 SOD methods in recent years, including PoolNet (Liu et al., 2019), PoolNet+ (Liu et al., 2022), F3Net (Wei et al., 2020a), LDF (Wei et al., 2020b), CII (Liu et al., 2021a), VST (Liu et al., 2021b), PSGNet (Yang

et al., 2021), PDRNet (Zhang et al., 2022), EDN (Wu et al., 2022), MMFT (Zhao et al., 2022), ICON (Zhuge et al., 2022), MENet (Wang et al., 2023), BBRF (Ma et al., 2023), HVPNet (Liu et al., 2020), SAMNet (Liu et al., 2021c), LSOD-P2T (Thu et al., 2024), ADMNet (Zhou et al., 2024) and CTDNet (Zhao et al., 2021), SOLNet (Li et al., 2025). Among these methods, HVPNet, SAMNet, CII, CTDNet, EDN, PoolNet+, LSOD-P2T, SOLNet and ADMNet+ are considered lightweight. Notably, VST and ICON-S are ViT architectures as in Fig. 2(b) and BBRF is a hybrid dual encoder architecture as in Fig. 2(c). SOLNet is a lightweight RSI-SOD method for remote sensing images.

Quantitative evaluation. As shown in Fig. 6, our method corresponds to a higher position of the PR curve, indicating better performance of the proposed model. Table 1 shows that the max-F, MAE and S-measure of our method on five mainstream SOD datasets are better than the existing lightweight methods. Meanwhile, our method achieves better performance with less computational cost compared to most heavyweight methods in recent years. Specifically, our XS version reduces the MAE by 0.015, 0.017, 0.013, 0.010, and 0.007 on the five datasets, respectively, compared to PoolNet-M+ with the same parameter size. Compared to its similar performance of heavyweight method MENet, our HASNet-S version reduces 20.9 million and 93.9 GFLOPs in the number of parameters and GFLOPs, respectively. Compared to the recently released method, our approach achieves superior detection results using only a quarter of the parameters required by the P2T version of LSOD. SOLNet is designed for the RSI-SOD task and seems to perform poorly after training on the RGB SOD dataset. Although our method is slightly more computationally expensive than ADMNet, it is far superior in performance. In addition, Fig. 7 presents the E-measure radar plots for all lightweight models, showing that our approach consistently outperforms others by wrapping well around them. Our method offers a significant performance advantage over the CNN-only architecture. When compared to methods based on hybrid ViT and dual-encoder architectures, our approach achieves a 10 to 20-fold reduction in parameter size and computational cost, with only a small performance gap. Based on the above observations, we conclude that our method achieves a better balance between segmentation accuracy and computational cost compared to other methods.

Qualitative evaluation. The qualitative comparison of our method with SOTA lightweight and heavyweight models is presented in Fig. 8. In the figure, areas of flawed segmentation in the saliency map are highlighted with red wireframes. Macroscopically, both of our models deliver image quality that surpasses existing lightweight models and rivals some heavyweight models. In more detail, the higher maximum feature resolution of our model performs well on fine-grained samples, as illustrated in rows 1 ~ 3. Our method also excels at segmenting salient objects in complex, multi-target scenes, shown in rows 4 ~ 5. Additionally, for large building samples displayed in rows 6 ~ 7, our method achieves complete segmentation of salient objects. Overall, our model shows superior effectiveness in various scenarios compared to other architectures.

4.5. Ablation study

Firstly, we conduct ablation experiments to evaluate the effectiveness of HAS. Secondly, we validate the effectiveness of the proposed LIRB and GIRB modules. Finally, we examine additional factors influencing the number of parameters, computational cost, and performance of HASNet. All experiments were performed on DUT-OMRON, which is the largest and most stable dataset.

Effectiveness of all proposed components. To demonstrate the effectiveness of our proposed components, we gradually extended the HAS, GIRB, LIRB and SMC from the baseline, and the results are shown in Table 2. We can see that the performance gradually improves with each of the proposed components. Optimal results are achieved on the DUT-OMRON dataset when all components are added, while the cost

¹ https://github.com/ArcherFMY/sal_eval_toolbox

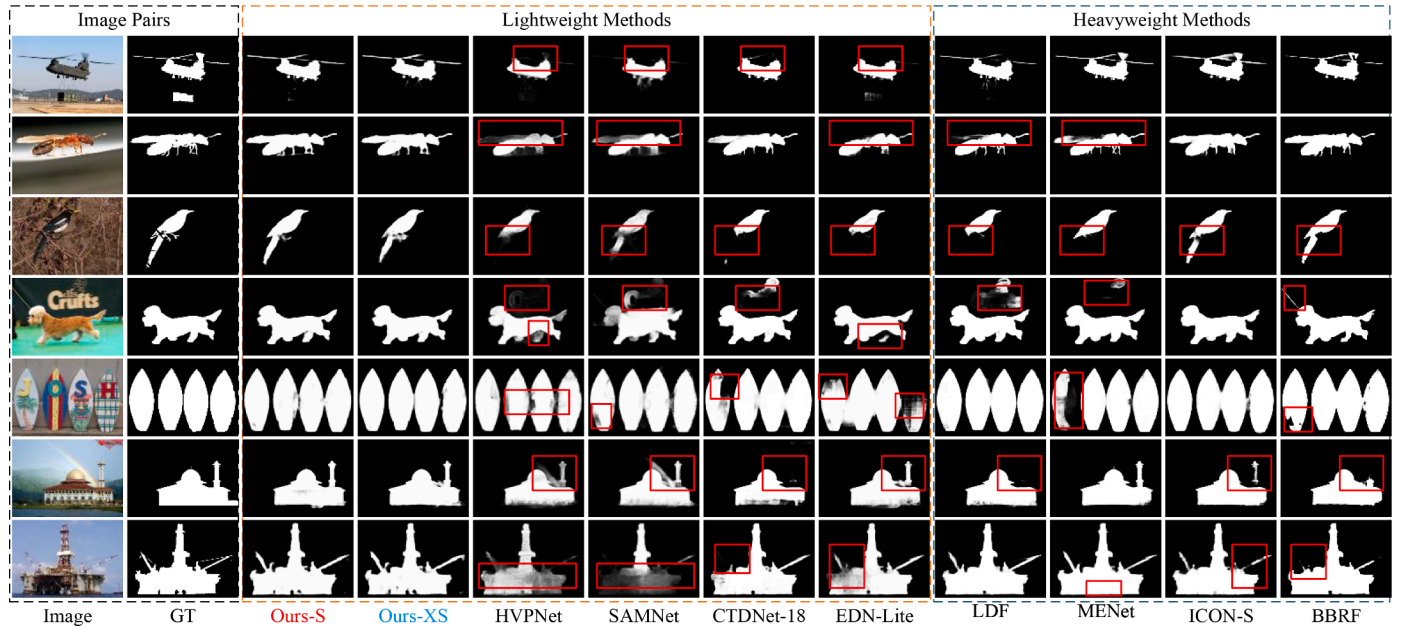


Fig. 8. Visual comparison of our lightweight models and some SOTA models. Our methods are colored red and blue.

Table 2

Analysis of the effectiveness of all proposed components. The best scores are **bold**.

No.	Baseline	HAS	GIRB	LIRB	SMC	Parameters (M)	FLOPs (G)	DUT-OMRON		
								$F_{\beta}^{\max} \uparrow$	MAE↓	$S_a \uparrow$
1	✓					2.92	0.32	0.826	0.053	0.850
2	✓	✓				3.04	0.96	0.833	0.052	0.854
3	✓	✓	✓			3.15	0.95	0.831	0.051	0.853
4	✓	✓	✓	✓		3.11	0.77	0.834	0.050	0.855
5	✓	✓	✓	✓	✓	3.03	0.54	0.835	0.049	0.856

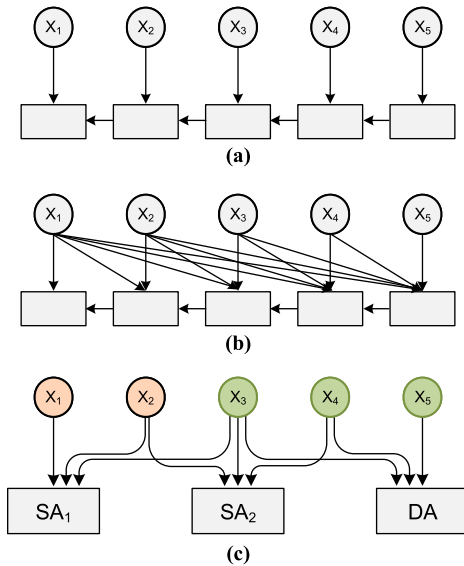


Fig. 9. Different aggregation strategy structures. (a) Structure of the cascade aggregation strategy. (b) Structure of the dense connection strategy. (c) Structure of our hybrid aggregation strategy.

compared to the baseline is only an additional 0.11 M parameter size and 0.22 GFLOPs.

Effectiveness of HAS. To demonstrate the effectiveness of HAS, we conduct both qualitative and quantitative experiments. In Table 3, (a)-

(c) present the experimental results of the corresponding aggregation strategies shown in Fig. 9. Experiments indicate that the cascade aggregation strategy in (a) fails to achieve satisfactory results. The densely connected aggregation strategy in (b) shows slight improvement over (a) but incurs a high computational cost and increases the number of parameters. The HAS strategy in (c) not only achieves the best performance but also demonstrates high efficiency. In addition, we conduct ablation experiments on module selection for HAS, focusing on cross-architecture feature aggregation, as shown in Table 4. The aggregation of (X_4, X_2) , (X_3, X_2) , and (X_3, X_1) in HAS all involve cross-architecture aggregations. We sequentially switch the aggregation module from LIRB to GIRB, demonstrating that LIRB provides higher performance and reduces computational cost compared to GIRB. This is because high-resolution features with a low number of channels are less suitable for attention mechanism computation. To further analyze the role of HAS in grouping and aggregating features from different architectures, we perform a Grad-CAM visualization of its different aggregation steps, as shown in Fig. 10. The visualized heat maps reveal that the aggregation of HAS in the transformer features focuses more on discriminative locations within the object, while the aggregation in the CNNs features covers the object more comprehensively. Without HAS, the model misdirects attention to incorrect locations during the aggregation of transformer features, which then affects the aggregation of CNNs features. The above experiments demonstrate that the HAS we designed is both necessary and effective for feature aggregation across different architectures.

Ablation analysis of LIRB. The efficiency ablation experiment for LIRB is presented in Table 5. The first row of the table shows the use of conventional IRBs as a baseline, replacing all LIRBs in our approach. The

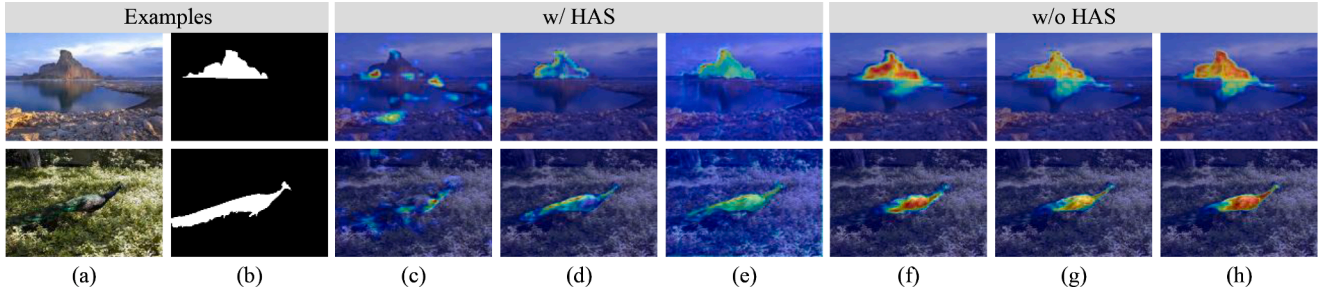


Fig. 10. Grad-CAM visualisation analysis of HAS. (a) Image; (b) Ground truth; (c)-(e) and (f)-(h) CAM plots from deep to shallow with and without the HAS, respectively.

Table 3

Effectiveness analysis of the proposed aggregation architecture. The best scores are **bold**.

No.	Strategies	Parameters (M)	FLOPs (G)	DUT-OMRON		
				$F_{\beta}^{\max} \uparrow$	MAE↓	$S_a \uparrow$
1	a	2.92	0.32	0.829	0.053	0.849
2	b	5.73	5.02	0.831	0.053	0.853
3	c	3.03	0.54	0.835	0.049	0.856

Table 4

Effectiveness analysis of using different IRB modules for cross-architecture aggregation. C_{i-j} denotes the cross-architecture aggregation between the i th layer feature and the j th layer feature. The best scores are **bold**.

No.	Strategies	Parameters (M)	FLOPs (G)	DUT-OMRON		
				$F_{\beta}^{\max} \uparrow$	MAE↓	$S_a \uparrow$
1	C_{4-2}	3.05	0.59	0.829	0.054	0.850
2	$C_{4-2,3-2}$	3.06	0.63	0.830	0.053	0.852
3	$C_{4-2,3-2,3-1}$	3.07	0.71	0.832	0.051	0.853
4	HAS	3.03	0.54	0.835	0.049	0.856

Table 5

Ablation analysis of LIRB. “E” and “S” denote the expansion stage and shrinkage stage, respectively. Parameters and GFLOPs represent efficiency metrics for individual module. The best scores are **bold**.

No.	Strategies	Parameters (M)	FLOPs (G)	DUT-OMRON		
				$F_{\beta}^{\max} \uparrow$	MAE↓	$S_a \uparrow$
1	IRB	0.0103	0.041	0.831	0.051	0.853
2	+ E	0.0080	0.032	0.834	0.050	0.853
3	+ S	0.0080	0.032	0.832	0.051	0.854
4	+ E + S	0.0057	0.022	0.834	0.050	0.855

second and third rows represent the experimental results of replacing the 1×1 convolution in the IRB with an expansion or shrinkage stage alone for lightweighting, respectively. The last row shows the experimental results of LIRB with both expansion and shrinkage stages. As can be seen in the results, the parameter size and computational cost of the model are decreasing as we improve the two 1×1 convolutional parts of the IRB. In addition, we examine the impact of the weighted summation approach during the shrinkage stage on LIRB’s performance, as shown in Table 6. The results indicate that our proposed weighted summation of adjacent channels during the shrinkage stage positively impacts feature aggregation.

Ablation analysis of GIRB. We conduct ablation experiments on the efficient global operation in GIRB, with the results shown in Table 7. The first row shows the results obtained by using conventional IRB for all aggregations of the feature $F_{3 \sim 5}$. The second row shows the results obtained by replacing the IRB with a cross-scale attention mechanism. Row 3 ~ 5 show the performance of our method using GIRB with differ-

Table 6

Ablation analysis of shrinkage stage. Parameters and GFLOPs represent efficiency metrics for individual module. CW denotes channel weighting, HCW denotes half channel weighting, and ACW denotes adjacent channel weighting. The best scores are **bold**.

No.	Strategies	Parameters (M)	GFLOPs (G)	DUT-OMRON		
				$F_{\beta}^{\max} \uparrow$	MAE↓	$S_a \uparrow$
1	w/o CW	0.0057	0.022	0.833	0.052	0.854
2	w/ HCW	0.0057	0.022	0.831	0.050	0.853
3	w/ ACW	0.0057	0.022	0.834	0.050	0.855

Table 7

Ablation analysis of GIRB. “C” denotes the cross-scale attention operation, and “ F_i ” represents the efficient global operation with a folding window size of i . Parameters and GFLOPs represent efficiency metrics for individual module. The best scores are **bold**.

No.	Strategies	Parameters (M)	FLOPs (G)	DUT-OMRON		
				$F_{\beta}^{\max} \uparrow$	MAE↓	$S_a \uparrow$
1	IRB	0.106	0.0270	0.828	0.053	0.851
2	+ C	1.161	0.6709	0.829	0.052	0.850
3	+ F_2	0.236	0.1205	0.830	0.051	0.852
4	+ F_4	0.236	0.1199	0.831	0.051	0.853
5	+ F_8	0.236	0.1197	0.835	0.049	0.856

Table 8

Effectiveness analysis of expand ratio. The best scores are **bold**.

No.	Expand ratio	Parameters (M)	FLOPs (G)	DUT-OMRON		
				$F_{\beta}^{\max} \uparrow$	MAE↓	$S_a \uparrow$
1	2	3.03	0.54	0.835	0.049	0.856
2	4	3.34	1.22	0.832	0.052	0.854

ent folding window sizes (8, 4, and 2). The results indicate that while a conventional IRB minimizes model parameter size and computational cost, it significantly reduces performance. Employing multi-head cross-attention considerably increases model complexity without substantial performance gains. In contrast, our proposed GIRB achieves optimal performance while maintaining low model complexity. Furthermore, high-level features effectively complement low-level features when the collapsed window size in efficient global operations is set to 8. We also conducted ablation experiments on the expansion ratio in LIRB and GIRB, as shown in Table 8. When the expansion ratio is reduced to 2, the parameter size and computational cost of HASNet are significantly optimized.

Effectiveness of SMC. The role of SMC is to fuse two same-scale features at the end of each aggregation stage. We conducted ablation experiments on SMC, as shown in Table 9. The CBP module with 3×3 convolution achieves strong results but imposes a high computational burden. Although 3×3 depth-wise convolution and LIRB are more resource-efficient, they lead to performance degradation. Our proposed SMC module effectively captures spatial and channel information by employ-

Table 9Effectiveness analysis of SMC. The best scores are **bold**.

No.	Modules	Parameters (M)	FLOPs (G)	DUT-OMRON		
				$F_{\beta}^{\max} \uparrow$	MAE $\downarrow$	$S_a \uparrow$
1	CBP _{3x3}	3.11	0.77	0.834	0.050	0.855
2	DWConv _{3x3}	3.01	0.48	0.829	0.053	0.851
3	LIRB	3.02	0.52	0.829	0.051	0.851
4	SMC	3.03	0.54	0.835	0.049	0.856

Table 10Effectiveness analysis of Backbone network. The best scores are **bold**.

No.	Backbone	Parameters (M)	FLOPs (G)	DUT-OMRON		
				$F_{\beta}^{\max} \uparrow$	MAE $\downarrow$	$S_a \uparrow$
1	ResNet-18	12.46	3.22	0.792	0.060	0.821
2	MobileNetV2	1.69	0.51	0.789	0.063	0.819
3	MobileViT-XS	3.00	0.50	0.835	0.049	0.856

Table 11Ablation analysis of parameter balancing in loss functions. $\lambda_{1\sim4}$ denote the weight parameters of $L_{1\sim4}$, respectively. The best scores are **bold**.

No.	λ_1	λ_2	λ_3	λ_4	DUT-OMRON		
					$F_{\beta}^{\max} \uparrow$	MAE $\downarrow$	$S_a \uparrow$
1	1.0	0.5	0.5	0.5	0.830	0.052	0.852
2	1.0	1.0	0.5	0.5	0.832	0.051	0.854
3	1.0	1.0	1.0	0.5	0.830	0.052	0.851
4	1.0	0.9	0.8	0.7	0.832	0.051	0.853
5	1.0	1.0	1.0	1.0	0.835	0.049	0.856

ing multiple dilation ratios in dilated convolutions, balancing computational resource consumption and performance.

Effectiveness analysis of Backbone network. We select a commonly used lightweight backbone network for ablation experiments on HASNet, as shown in Table 10. The non-hybrid encoder architectures, ResNet-18 (He et al., 2016) and MobileNetV2 (Sandler et al., 2018), are less suitable for our HAS feature aggregation strategy, resulting in poorer performance due to their reliance on single convolutional features. In contrast, MobileViT's hybrid architecture encoder, which better adapts to our proposed HAS, achieves optimal performance.

Effectiveness analysis of input sizes. We analyze the impact of common input image sizes ($224 \times 224 \sim 384 \times 384$) on the accuracy and computational cost of our method, as shown by the purple solid and dashed lines in Fig. 11. As expected, the computational cost increases with the input size. The MAE reaches its minimum at an input size of 256×256 and begins to rise when the input size exceeds 256×256 . This could be due to the small parameter size of the model, which struggles to handle higher resolution images. We perform the same test for EDN-Lite, which has a similar computational cost, as shown in the green solid and dashed parts in Fig. 11. The results show that EDN-Lite achieves the minimum MAE with an input size of 320×320 , suggesting that HASNet utilizes a more efficient input resolution to achieve better detection.

Loss function parameter balance analysis. As shown in Table 11, we analyse the four deep supervisory weights in the loss function. In our experiments, we set the loss weight of the main output to 1 and successively reduced the loss weights of the other branch outputs. The results show that the performance of the model gradually improves the more weight that is given to each branch. The best results are achieved when all weights are set to 1.

5. Limitations

Although our method achieves high accuracy with efficiency, there are still areas for improvement. For instance, while our method requires very low GFLOPs, the FPS of the transformer-based architecture lags

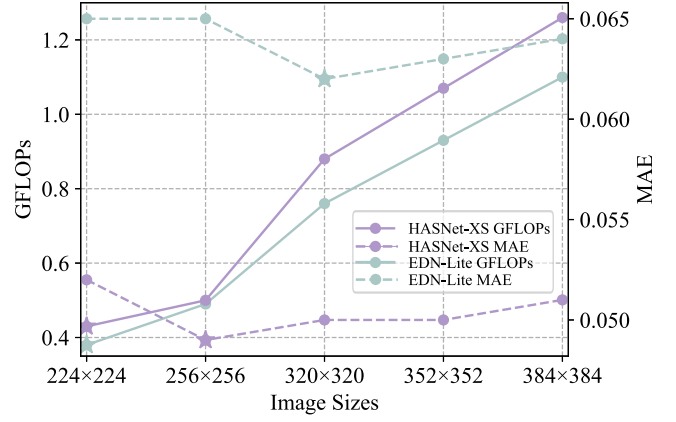


Fig. 11. Effect of different input sizes for the models in MAE and GFLOPs. All results were obtained by testing on the DUT-OMRON dataset.

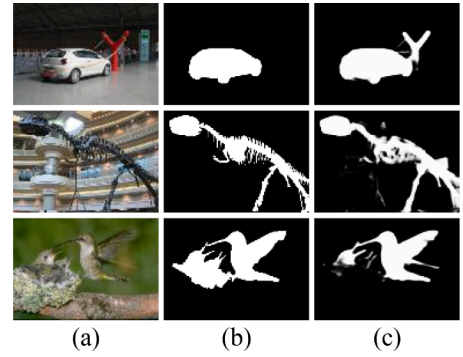


Fig. 12. Some failure cases of our method in extreme environments. (a) Image; (b) Ground truth; (c) Ours.

behind that of CNNs. This issue can be addressed by optimizing the transformer implementation.

In addition, we observe that HASNet encounters detection failures in certain specific scenarios, as shown in Fig. 12. The overly prominent color of the background affects the detection of salient objects (e.g., 1st row). In images with large objects containing rich details, some finer details are lost in the saliency map (e.g., 2nd row). Additionally, multi-target images with partially obscured objects are not fully segmented (e.g., 3rd row). Our future work can further optimize lightweight SOD to address these specific scenarios, such as filtering out overly bright, non-salient objects and incorporating multi-objective scene datasets. Additionally, the lack of information interaction caused by the folding operation in GIRB is expected to be mitigated by the establishment of a low-cost fusion method within the folding window.

6. Conclusion

In this paper, we design a lightweight SOD network based on a hybrid architecture. To achieve optimal segmentation with limited resources, we propose a hybrid aggregation strategy for combining hybrid features. In HAS, we design LIRB and GIRB to aggregate different types of features at various stages. LIRB enhances two 1×1 convolutions in the expansion and shrinkage stages based on IRB, significantly reducing computational and storage resource usage. In GIRB, we introduce an efficient global operation that fully utilizes channel semantic information within transformer features, achieving effective detection with minimal cost. By combining HAS, LIRB and GIRB, we exploit the performance potential of lightweight SOD and offer two model versions: HASNet-XS (3.0M, 603 FPS) and HASNet-S (6.9M, 492 FPS), designed for practical applications in different environments. Extensive comparative experiments and ablation studies demonstrate the effectiveness of our proposed HASNet,

achieving a favorable trade-off between efficiency and performance in lightweight SOD.

CRedit authorship contribution statement

Jianhua Ma: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Mingfeng Jiang:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization; **Xian Fang:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis; **Jiatong Chen:** Writing – review & editing, Validation; **Yaming Wang:** Writing – review & editing, Funding acquisition; **Guang Yang:** Writing – review & editing, Funding acquisition.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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